

Ehsan Barkhordar

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📄 [Stack Overflow](https://stackoverflow.com/users/1041444/ebarkhordar)

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🎓 [Google Scholar](https://scholar.google.com/citations?user=ebarkhordar)

Education

Koç University (Ranked #1 in Turkey, QS 2026)

M.Sc. in Computer Science, GPA: 3.8/4.0

Feb 2023 – Feb 2026

Istanbul, Turkey

- Advisors: Prof. Gözde Gül Şahin (primary), Assoc. Prof. Erdem Yörük (co-advisor).

Amirkabir University of Technology (Ranked #3 in Iran, QS 2026)

M.Sc. in Computer Science, GPA: 15.15/20 (3.1/4.0)

Sep 2018 – Mar 2021

Tehran, Iran

Amirkabir University of Technology (Ranked #3 in Iran, QS 2026)

B.Sc. in Computer Science, GPA: 14.79/20 (3.0/4.0)

Sep 2013 – Apr 2018

Tehran, Iran

Publications

Are Non-English Papers Reviewed Fairly? Language-of-Study Bias in NLP Peer Reviews

(EMNLP 2026)

- Ehsan Barkhordar, Abdulfattah Safa, Verena Blaschke, Erika Lombart, Marie-Catherine de Marneffe, Gözde Gül Şahin. *Proceedings of EMNLP 2026 (Main Conference)*, 2026, Budapest, Hungary. To appear.

Assessing the Predictive Power of Social Media Data-Fed Large Language Models on Vote Preference

- Ehsan Barkhordar, Sukru Atsizelti. *Companion Publication of the 16th ACM Web Science Conference*, 2024, Stuttgart, Germany.

Why the Unexpected? Dissecting the Political and Economic Bias in Persian Small and Large Language Models

- Ehsan Barkhordar, Surendrabikram Thapa, Ashwarya Maratha, Usman Naseem. *Proceedings of the 3rd Annual Meeting of SIGUL @ LREC-COLING*, 2024, Torino, Italy.

Team Curie at HSD-2Lang 2024: Hate Speech Detection in Turkish and Arabic Tweets using BERT-based Models

- Ehsan Barkhordar, Işık Topçu, Ali Hürriyetoğlu. *Proceedings of the 7th Workshop on CASE*, 2024, St. Julians, Malta.

Which Side Are You On? Investigating Politico-Economic Bias in Nepali Language Models

- Surendrabikram Thapa, Kritesh Rauniyar, Ehsan Barkhordar, Hariram Veeramani, Usman Naseem. *Proceedings of ALTA*, 2024, Canberra, Australia.

Clustering of Bank Customers using LSTM-based Encoder-Decoder and Dynamic Time Warping

- Ehsan Barkhordar, Mohammad Hassan Shiralil-Shahreza, Hamid Reza Sadeghi. *arXiv preprint arXiv:2110.11769*, 2021.

Teaching Experience

Koç University

Fall 2024 – Spring 2025

Teaching Assistant – COMP 302: Software Engineering (2 consecutive terms)

Istanbul, Turkey

- Facilitated lectures and labs on software engineering practices including testing, CI/CD, and agile methods.
- Designed assignment rubrics, evaluated semester projects, and organized discussion sections.
- Provided technical mentorship to 60+ students on system design, version control (Git), and collaborative workflows.

Koç University

Spring 2024

Teaching Assistant – COMP 100: Introduction to CS and Programming

Istanbul, Turkey

- Supported an introductory Python programming course with 200+ students enrolled.
- Graded weekly assignments, organized lab sessions, and assisted students in debugging.

Koç University

Fall 2023

Teaching Assistant – COMP 491: Senior Design Project

Istanbul, Turkey

- Advised senior students on project design, implementation, and research methodology.
- Evaluated technical documentation and presentations, and provided structured feedback.

Koç University

Spring 2023

Teaching Assistant – COMP 202: Data Structures and Algorithms in Java

Istanbul, Turkey

- Led weekly lab/tutorial sessions for 50+ undergraduate students on data structures and Java programming.
- Held office hours, graded assignments/projects, and provided one-on-one academic support.

Research & Work Experience

Koç University

Feb 2023 – Feb 2026

Researcher in NLP & Computational Social Science

Istanbul, Turkey

- Led a multimodal hate speech detection project integrating textual and visual modalities.
- Authored three first-author papers in computational social science.
- Reviewed 100+ papers for ARR, EMNLP, ICLR, and related venues.
- Produced 65+ weekly progress reports for collaborative research tracking.
- Trained and fine-tuned transformer-based models (BERT, RoBERTa, GPT) for NER and related tasks.
- Supervised two research interns, providing mentorship and task delegation.

Enlighty

Sep 2023 – Feb 2026

Technical Lead

Remote

- Set technical direction for an AI consumer-intelligence platform (emotion detection, consumer journey, and brand health analytics) spanning a FastAPI/ClickHouse backend, two Nuxt 4 frontends, and an LLM assistant, supporting a team of 12 engineers.
- Built the AI infrastructure behind *Enlighter*, an agentic analytics assistant: LangChain/OpenAI tool-calling over 83 analytics endpoints, WebSocket response streaming, templated prompt management, and automated PDF report generation.
- Cut API latency by replacing a singleton ClickHouse client with a thread-safe connection pool, moving blocking queries off the async event loop, and eliminating N+1 JWT lookups.
- Containerized all four services and built GitHub Actions pipelines deploying to isolated staging, demo, and production environments, with automated cache flushing and post-deploy warm-up.
- Added multi-region ClickHouse routing and led recurring code audits and refactors, including splitting a 692-line auth router into five focused modules and standardizing configuration and structured logging.

Pishro Net Energy

Aug 2022 – Jan 2023

Machine Learning Engineer

Tehran, Iran

- Designed and deployed end-to-end ML pipelines for real-time energy monitoring across 1,000+ sensor streams (10 buildings), reducing forecasting error by 21%.
- Created data-driven dashboards for 150+ KPIs to support facility managers in decision-making.

Sokan Platform

Apr 2021 – Feb 2022

Machine Learning Engineer

Tehran, Iran

- Built predictive models for customer analytics, including churn prediction (82% accuracy) and cohort-based sales insights.
- Optimized large-scale data pipelines (1B+ records) using TensorFlow, Kubeflow, and Mini-batch K-means.
- Applied FP-Growth to optimize product bundling strategy, improving cross-sell opportunities and data-driven merchandising decisions.

Sokan Platform

Sep 2020 – Apr 2021

Senior Software Engineer

Tehran, Iran

- Implemented asynchronous task management in Django with Celery, reducing computational overhead by 40%.
- Accelerated analytics retrieval speed 5x via ClickHouse migration.
- Delivered 50+ GraphQL APIs using Django Graphene, reducing frontend data-fetching overhead by 35% and consolidating multiple REST endpoints.
- Architected and deployed automated CI/CD pipelines with Docker and GitLab, cutting deployment time by 60% and enabling 3x more frequent releases.

Bale Messenger

Jan 2018 – Feb 2020

Software Engineer

Tehran, Iran

- Contributed to the development of a cross-platform messaging application with 8M+ users.
- Created a Python-based bot SDK adopted by 200+ developers, powering 500+ bots and broadening accessibility for both developers and non-developers.
- Participated in migration to Golang microservices, achieving >7x performance gains.
- Implemented monitoring pipelines with Prometheus and Grafana to track 20+ key system metrics.

Skills

Programming: Python (Advanced, OOP, Async), SQL (Analytical & OLTP), Bash/Shell

Python Frameworks: PyTorch, Hugging Face (Transformers, Datasets, Accelerate, PEFT), Scikit-learn, Pandas, NumPy

GenAI & NLP: LLMs (Fine-tuning, RAG), LangChain, Embeddings & Retrieval, Prompt Engineering, LLM Reasoning, LLM Evaluation, Text Classification, Multilingual NLP

Research Tools: LaTeX (ACL/IEEE/ACM styles), Overleaf, Git/GitHub, Jupyter, MLflow/W&B, Experiment Tracking, Reproducibility Pipelines

Databases & Analytics: PostgreSQL (Production, Performance Tuning), MongoDB, ClickHouse

Backend & APIs: FastAPI, REST APIs, GraphQL, WebSockets, Celery, Redis, Auth (JWT/OAuth2), HTML/CSS, JavaScript, TypeScript/Nuxt

Dev Tools: Git/GitHub, Docker, GitHub Actions (CI/CD), Jupyter, Cursor

Languages

Persian: Native

English: Fluent – TOEFL iBT 103/120 (R:26, L:28, S:25, W:24), Dec 2024

Turkish: Beginner

Selected Projects

LLM Collusion and Identifiability in Code Generation | May 2025 – Present

- Investigates whether large language models can recognize or attribute code authored by themselves or other LLMs.
- Implemented three attribution tasks: Self-Recognition, Target Identification, and Full Attribution.
- Evaluated across multiple benchmarks (MBPP, HumanEval, DS-1000) with code generated by GPT-5, Claude, Gemini, Grok, and DeepSeek.
- Fine-tuned lightweight judge models to identify model-specific code signatures, highlighting risks of collusion and leakage in code evaluation.

Multimodal Framework for Detecting Implicit Hate Speech on Social Media | Sept 2024 – Apr 2025

- Designed a hybrid framework combining text encoders (BERT, LLaMA) with vision backbones (Swin Transformer, ViT) to detect subtle hate speech.
- Improved robustness via focal loss, progressive unfreezing, and adversarial training.
- Demonstrated superior performance on multimodal hate speech datasets, addressing explainability and ethical implications.

Named Entity Recognition in Turkish Tweets (Yildirim Platform) | Jun 2024 – Sept 2024

- Developed an annotation platform (Yildirim) for labeling named entities in Turkish tweets.
- Annotated 5,000 tweets from a 700M+ tweet dataset to improve NER performance on low-resource, informal language.

Procedural Text Understanding via Python Code Representation | Jul 2023 – Sept 2023

- Explored Python code representation as an intermediate reasoning layer for LLMs.
- Code-based prompting improved accuracy from 68.29% to 86.76%.

Intent Prediction and Slot Filling with Alexa MASSIVE Dataset | Feb 2022 – Apr 2022

- Participated in the Massively Multilingual NLU 2022 shared task.
- Trained multilingual intent and slot-filling models on the Alexa MASSIVE dataset using PyTorch.

Shilling Attack Detection in Recommender Systems | May 2020

- Investigated detection of profile injection (shilling) attacks in recommender systems.
- Extracted RDMA, WDA, and TF-IDF features; evaluated multiple classifiers (KNN, Random Forest, SVM).

Honors and Awards

2nd Rank – Hate Speech Detection in Turkish and Arabic Tweets (HSD-2Lang) 2024

Invited Participant – Lisbon Machine Learning School (LxMLS 2023), Lisbon, Portugal 2023

One of 5 Selected Teams – Fusion Startup Event (sponsored by divar.ir) 2022

28th Place / 1608 Participants – Django/Python Code Cup, quera.ir 2020

2nd Team – Hackathon, Amirkabir University 2019

Winner – BOT-CUP, Sharif University of Technology 2018